

File permissions in Linux

Project description

[This project demonstrates the use of Linux commands to manage file and directory permissions. I used commands such as `ls -la` and `chmod` to inspect and modify permissions, including permissions for hidden files and directories. The goal was to ensure that files and directories are accessible only to authorized users.]

Check file and directory details

```
[
  r5gc7twqswqqidqsbmv2svqqz4aqp33rot5q2govfrvtoft73kuq.us-central1-c.resources.bumper-boats-00.services.qwiklabs.com
  researcher2@74b5flad0ac5:~$ cd projects
  researcher2@74b5flad0ac5:~/projects$ ls -l
  total 20
  drwx--x--- 2 researcher2 research_team 4096 Aug 25 07:50 drafts
  -rw-rw-rw- 1 researcher2 research_team 46 Aug 25 07:50 project_k.txt
  -rw-r----- 1 researcher2 research_team 46 Aug 25 07:50 project_m.txt
  -rw-rw-r-- 1 researcher2 research_team 46 Aug 25 07:50 project_r.txt
  -rw-rw-r-- 1 researcher2 research_team 46 Aug 25 07:50 project_t.txt
  researcher2@74b5flad0ac5:~/projects$ ls -la
  total 32
  drwxr-xr-x 3 researcher2 research_team 4096 Aug 25 07:50 .
  drwxr-xr-x 3 researcher2 research_team 4096 Aug 25 08:05 ..
  -rw--w---- 1 researcher2 research_team 46 Aug 25 07:50 .project_x.txt
  drwx--x--- 2 researcher2 research_team 4096 Aug 25 07:50 drafts
  -rw-rw-rw- 1 researcher2 research_team 46 Aug 25 07:50 project_k.txt
  -rw-r----- 1 researcher2 research_team 46 Aug 25 07:50 project_m.txt
  -rw-rw-r-- 1 researcher2 research_team 46 Aug 25 07:50 project_r.txt
  -rw-rw-r-- 1 researcher2 research_team 46 Aug 25 07:50 project_t.txt
  researcher2@74b5flad0ac5:~/projects$
```

ls -la

File/Directory	Current Permissions
drafts	drwxr-x---

```
.project_x.  -rwx-w----
txt

project_k.t  -rw-rw-rw-
xt

project_m.t  -rw-r-----
xt

project_r.t  -rw-rw-r--
xt

project_t.t  -rw-rw-r--
xt
]
```

Describe the permissions string

[The 10-character permission string `-rw-rw-r--` represents the file type and the access permissions assigned to the user, group, and others. The first character `-` indicates that the item is a regular file. The next three characters `rw-` show that the user has read and write permissions. The following three characters `rw-` show that the group has read and write permissions. The final three characters `r--` show that others have read-only permission and cannot write or execute the file.]

Change file permissions

```
[
-rw-rw-r-- 1 researcher2 research_team 46 Aug 25 07:50 project_r.txt
researcher2@74b5flad0ac5:~/projects$ chmod o-w project_k.txt
researcher2@74b5flad0ac5:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 25 07:50 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 25 08:05 ..
-rw-w---- 1 researcher2 research_team 46 Aug 25 07:50 .project_x.txt
drwx-x--- 2 researcher2 research_team 4096 Aug 25 07:50 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Aug 25 07:50 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Aug 25 07:50 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 25 07:50 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 25 07:50 project_t.txt
researcher2@74b5flad0ac5:~/projects$ chmod g-r project_m.txt
researcher2@74b5flad0ac5:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 25 07:50 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 25 08:05 ..
-rw-w---- 1 researcher2 research_team 46 Aug 25 07:50 .project_x.txt
drwx-x--- 2 researcher2 research_team 4096 Aug 25 07:50 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Aug 25 07:50 project_k.txt
-rw----- 1 researcher2 research_team 46 Aug 25 07:50 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 25 07:50 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 25 07:50 project_t.txt
researcher2@74b5flad0ac5:~/projects$ ]
```

The `chmod o-w project_k.txt` command removes write permission for other users from

`project_k.txt`. I then used `ls -la` to verify that the permission was changed from `-rw-rw-rw-` to `-rw-rw-r--`.]

Change file permissions on a hidden file

[

```
researcher2@74b5flad0ac5:~/projects$ chmod g-w .project_x.txt
researcher2@74b5flad0ac5:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 25 07:50 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 25 08:05 ..
-r--r----- 1 researcher2 research_team  46 Aug 25 07:50 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug 25 07:50 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Aug 25 07:50 project_k.txt
-rw----- 1 researcher2 research_team  46 Aug 25 07:50 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 25 07:50 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 25 07:50 project_t.txt
researcher2@74b5flad0ac5:~/projects$
```

The `chmod g-w .project_x.txt` command removes write permission from the group for the hidden file `.project_x.txt`. After this change, the file does not have write permission for the group, while the user and group can retain read access according to the required permissions.]

Change directory permissions

[

```
researcher2@74b5flad0ac5:~/projects$ cd drafts
researcher2@74b5flad0ac5:~/projects/drafts$
researcher2@74b5flad0ac5:~/projects/drafts$ ls -la
total 8
drwx--x--- 2 researcher2 research_team 4096 Aug 25 07:50 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 25 07:50 ..
researcher2@74b5flad0ac5:~/projects/drafts$ chmod g-x .
researcher2@74b5flad0ac5:~/projects/drafts$ ls -la
total 8
drwx----- 2 researcher2 research_team 4096 Aug 25 07:50 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 25 07:50 ..
researcher2@74b5flad0ac5:~/projects/drafts$
```

The `chmod g-x .` command removes execute permission from the group for the current `drafts` directory. The `ls -la` command was then used to verify the updated permissions. The output shows that the directory permissions were restricted so that only the owner, `researcher2`, retains full access.]

Summary

[In this activity, I used Linux commands to check and modify file and directory permissions. I used `ls -la` to view permissions, including hidden files, and `chmod` to remove unauthorized write and access permissions. These steps help protect organizational files and ensure that only authorized users can access sensitive information.]